What gridmicrotex draws

Each row is a LaTeX input and what the package makes of it, three to a line where they fit: accents, delimiters, operators, relations and symbols first, then further parts of LaTeX – units, equations, tables, diagrams, TikZ pictures, boxes and columns. A row is a test: it must be drawn with no warning. Where LaTeX has nothing to compare with, only the name is listed.

This PDF is rendered with this package.

Accents

`a'`	a'	`\tilde{a}`	$\tilde{a}$	`\mathring{g}`	$\mathring{g}$
`a''`	a''	`\widetilde{ac}`	$\widetilde{ac}$	`a^{\prime}`	$a^{\prime}$
`\utilde{AB}`	$\underset{\sim}{AB}$	`\acute{a}`	$\acute{a}$	`\vec{F}`	$\vec{F}$
`\bar{y}`	$\bar{y}$	`\breve{a}`	$\breve{a}$	`\check{a}`	$\check{a}$
`\dot{a}`	$\dot{a}$	`\overbrace{AB}`	$\overbrace{AB}$	`\ddot{a}`	$\ddot{a}$
`\underbrace{AB}`	$\underbrace{AB}$	`\dddot{a}`	$\dddot{a}$	`\overline{AB}`	$\overline{AB}$
`\overbracket{AB}`	$\overbracket{AB}$	`\ddddot{a}`	$\ddddot{a}$	`\underline{AB}`	$\underline{AB}$
`\underbracket{AB}`	$\underbracket{AB}$	`\grave{a}`	$\grave{a}$	`\underbar{X}`	$\underline{X}$
`\hat{\theta}`	$\hat{\theta}$	`\widehat{ac}`	$\widehat{ac}$	`\text{\'{a}}`	$\text{\'{a}}$
`\text{\~{a}}`	$\text{\~{a}}$	`\text{\.{a}}`	$\text{\.{a}}$	`\text{\`{a}}`	$\text{\`{a}}$
`\text{\={a}}`	$\text{\={a}}$	`\text{\"{a}}`	$\text{\"{a}}$	`\text{\v{a}}`	$\text{\v{a}}$
`\text{\^{a}}`	$\text{\^{a}}$	`\text{\u{a}}`	$\text{\u{a}}$	`\text{\r{a}}`	$\text{\r{a}}$
`\text{\H{o}}`	$\text{\H{o}}$				

`\overleftarrow{AB}`	$\overleftarrow{AB}$	`\overrightarrow{AB}`	$\overrightarrow{AB}$
`\underleftarrow{AB}`	$\underleftarrow{AB}$	`\underrightarrow{AB}`	$\underrightarrow{AB}$
`\overleftrightarrow{AB}`	$\overleftrightarrow{AB}$	`\underleftrightarrow{AB}`	$\underleftrightarrow{AB}$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\widecheck{ac}` `\overgroup{AB}` `\undergroup{AB}`
`\overlinesegment{AB}` `\underlinesegment{AB}` `\Overrightarrow{AB}`
`\overleftharpoon{ac}` `\overrightharpoon{ac}`

Delimiters

`(~)`	$(~)$	`\lparen~\rparen`	$\lparen~\rparen$	`\lceil~\rceil`	$\lceil~\rceil$
`\uparrow`	$\uparrow$	`[~]`	$[~]$	`\lbrack~\rbrack`	$\lbrack~\rbrack$
`\lfloor~\rfloor`	$\lfloor~\rfloor$	`\downarrow`	$\downarrow$	`\{ \}`	$\{~\}$
`\lbrace \rbrace`	$\lbrace~\rbrace$	`\updownarrow`	$\updownarrow$	`\langle~\rangle`	$\langle~\rangle$
`\lgroup~\rgroup`	$\lgroup~\rgroup$	`\Uparrow`	$\Uparrow$	`\vert`	$\vert$
`\Downarrow`	$\Downarrow$	`\Vert`	$\Vert$	`\Updownarrow`	$\Updownarrow$
`\lvert~\rvert`	$\lvert~\rvert$	`\lVert~\rVert`	$\lVert~\rVert$	`\backslash`	$\backslash$
`/`	$/$				

`\lmoustache \rmoustache`	$\lmoustache~\rmoustache$	`\ulcorner \urcorner`	$\ulcorner~\urcorner$
`\llcorner \lrcorner`	$\llcorner~\lrcorner$	`\llbracket~\rrbracket`	$\llbracket~\rrbracket$

`( \big( \Big( \bigg( \Bigg(` $(~\big(~\Big(~\bigg(~\Bigg($

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\lang~\rang` `\left\lt~\right\gt` `\lBrace~\rBrace`
`\left(\LARGE{AB}\right)`

Environments

```
\begin{matrix} a & b \\ c & d \end{matrix}
```

$$\begin{matrix} a & b \\ c & d \end{matrix}$$

```
\begin{array}{cc}a & b\\c & d\end{array}
```

$$\begin{array}{cc}a & b\\c & d\end{array}$$

```
\begin{pmatrix} a & b \\ c & d \end{pmatrix}
```

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

```
\begin{bmatrix} a & b \\ c & d \end{bmatrix}
```

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

```
\begin{vmatrix} a & b \\ c & d \end{vmatrix}
```

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

```
\begin{Vmatrix} a & b \\ c & d \end{Vmatrix}
```

$$\begin{Vmatrix} a & b \\ c & d \end{Vmatrix}$$

```
\begin{Bmatrix} a & b \\ c & d \end{Bmatrix}
```

$$\begin{Bmatrix} a & b \\ c & d \end{Bmatrix}$$

```
x = \begin{cases} a &\text{if } b \\ c &\text{if } d \end{cases}
```

$$x = \begin{cases} a &\text{if } b \\ c &\text{if } d \end{cases}$$

```
\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}
```

$$\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}$$

```
\sum_{\begin{subarray}{l} i\in\Lambda\\ 0<j<n\end{subarray}}
```

$$\sum_{\begin{subarray}{l} i\in\Lambda\\ 0<j<n\end{subarray}}$$

```
\begin{CD}A @>a>> B \\@VbVV @AAcA\\C @= D\end{CD}
```

$$\begin{CD}A @>a>> B \\@VbVV @AAcA\\C @= D\end{CD}$$

```
  \begin{equation}\begin{split}a
&=b+c\\&=e+f\end{split}\end{equation}
```

$$\begin{equation}\begin{split}a &=b+c\\&=e+f\end{split}\end{equation}$$

Environments (continued)

```
\begin{align} a&=b+c \\ d+e&=f \end{align}
```

$$a = b + c \tag{1}$$
$$d + e = f \tag{2}$$

```
\begin{gather} a=b \\ e=b+c \end{gather}
```

$$a = b \tag{3}$$
$$e = b + c \tag{4}$$

```
\begin{alignat}{2}10&x+&3&y=2\\3&x+&13&y=4\end{alignat}
```

$$10\,x + 3\,y = 2 \tag{5}$$
$$3\,x + 13\,y = 4 \tag{6}$$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\def\arraystretch{1.5}\begin{array}{c:c:c} a & b & c \\ \hline d & e & f \\ \hdashline g & h & i \end{array}
```

HTML

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\url{https://katex.org/}     \htmlId{bar}{x}              \htmlClass{foo}{x}
\htmlStyle{color: red;}{x}   \htmlData{foo=a, bar=b}{x}
\href{https://katex.org/}{\KaTeX}
```

Letters and Unicode

`\Gamma`	Γ	`\Delta`	Δ	`\Theta`	Θ
`\Lambda`	Λ	`\Xi`	Ξ	`\Pi`	Π
`\Sigma`	Σ	`\Upsilon`	Υ	`\Phi`	Φ
`\Psi`	Ψ	`\Omega`	Ω	`\varGamma`	Γ
`\varDelta`	Δ	`\varTheta`	Θ	`\varLambda`	Λ
`\varXi`	Ξ	`\varPi`	Π	`\varSigma`	Σ
`\varUpsilon`	Υ	`\varPhi`	Φ	`\varPsi`	Ψ
`\varOmega`	Ω	`\alpha`	α	`\beta`	β
`\gamma`	γ	`\delta`	δ	`\epsilon`	ϵ
`\zeta`	ζ	`\eta`	η	`\theta`	θ
`\iota`	ι	`\kappa`	κ	`\lambda`	λ
`\mu`	μ	`\nu`	ν	`\xi`	ξ
`\pi`	π	`\rho`	ρ	`\sigma`	σ
`\tau`	τ	`\upsilon`	υ	`\phi`	ϕ
`\chi`	χ	`\psi`	ψ	`\omega`	ω
`\varepsilon`	ε	`\varkappa`	ϰ	`\vartheta`	ϑ
`\varpi`	ϖ	`\varrho`	ϱ	`\varsigma`	ς
`\varphi`	φ	`\digamma`	ϝ	`\imath`	ı
`\nabla`	∇	`\Im`	ℑ	`\text{\OE}`	Œ
`\jmath`	ȷ	`\partial`	∂	`\wp`	℘
`\text{\o}`	ø	`\aleph`	ℵ	`\Game`	⅁
`\Bbbk`	𝕜	`\text{\O}`	Ø	`\Finv`	Ⅎ
`\text{\ss}`	ß	`\text{\aa}`	å	`\text{\i}`	ı

Letters and Unicode (continued)

`\beth`	ב	`\text{\AA}`	Å	`\text{\j}`	ȷ
`\gimel`	ג	`\ell`	ℓ	`\Re`	ℜ
`\text{\ae}`	æ	`\daleth`	ד	`\hbar`	ℏ
`\text{\AE}`	Æ	`\eth`	ð	`\hslash`	ℏ
`\text{\oe}`	œ				

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\Alpha      \Beta       \Epsilon
\Zeta       \Eta        \Iota
\Kappa      \Mu         \Nu
\Omicron    \Rho        \Tau
\Chi        \omicron    \thetasym
\Reals      \image      \weierp
\alef       \N          \Z
\alefsym    \cnums      \natnums
\Complex    \R          \real
\reals      \char"263a
```

Layout

Annotation

`\cancel{5}` $\cancel{5}$ `\bcancel{5}` $\bcancel{5}$ `\xcancel{ABC}` $\xcancel{ABC}$

`\not =` $\not =$ `\text{\sout{abc}}` ~~abc~~

`\overbrace{a+b+c}^{\text{note}}` $\overbrace{a+b+c}^{\text{note}}$ `\boxed{\pi=\frac c d}` $\boxed{\pi=\frac{c}{d}}$

`\text{\textcircled{a}}` ⓐ

```
\underbrace{a+b+c}_{\text{note}}
```

$$\underbrace{a+b+c}_{\text{note}}$$

```
\overbracket{a+b+c}^{\text{note}}
```

$$\overbracket{a+b+c}^{\text{note}}$$

```
\underbracket{a+b+c}_{\text{note}}
```

$$\underbracket{a+b+c}_{\text{note}}$$

```
\tag{hi} x+y^{2x}
```

$$x+y^{2x} \tag{hi}$$

```
\tag*{hi} x+y^{2x}
```

$$x+y^{2x} \tag*{hi}$$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\phase{30^\circ}   \angl{n}   \angln
```

Vertical Layout

`x_n`	x_n	`\stackrel{!}{=}`	$\stackrel{!}{=}$	`a \atop b`	$\begin{matrix}a\\b\end{matrix}$
`e^x`	e^x	`\overset{!}{=}`	$\overset{!}{=}$	`_u^o`	$\begin{matrix}o\\u\end{matrix}$
`\underset{!}{=}`	$\underset{!}{=}$				

`a\raisebox{0.25em}{$b$}c` $a^{b}c$ `\reflectbox{KaTeX}` XeTaK

`\sum_{\substack{0<i<m\\0<j<n}}` $\sum_{\substack{0<i<m\\0<j<n}}$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\mathreflectbox{x^2}` `\vcenter{x}`

Overlap and Spacing

`{=}\mathllap{/\,}` $=\!\!\!/$ `\mathrlap{\,/}{=}` $=\!\!\!/$

`\left(x^{\smash{2}}\right)` $\left(x^{\smash{2}}\right)$ `\sqrt{\smash[b]{y}}` $\sqrt{\smash[b]{y}}$

`\displaystyle\sum_{\mathclap{1\le i\le j\le n}} x_{ij}`

$$\displaystyle\sum_{\mathclap{1\le i\le j\le n}} x_{ij}$$

Logic and Set Theory

`\forall`	$\forall$	`\complement`	$\complement$	`\therefore`	$\therefore$
`\emptyset`	$\emptyset$	`\exists`	$\exists$	`\subset`	$\subset$
`\because`	$\because$	`\supset`	$\supset$	`\mapsto`	$\mapsto$
`\varnothing`	$\varnothing$	`\nexists`	$\nexists$	`\mid`	$\mid$
`\to`	$\to$	`\implies`	$\implies$	`\in`	$\in$
`\land`	$\land$	`\gets`	$\gets$	`\impliedby`	$\impliedby$
`\lor`	$\lor$	`\leftrightarrow`	$\leftrightarrow$	`\iff`	$\iff$
`\notin`	$\notin$	`\ni`	$\ni$	`\neg`	$\neg$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\empty` `\exist` `\isin`

`\notni` `\set{x\VERT x<5}`

`\Set{ x \VERT x<\frac 1 2 }`

Macros

`\def\foo{x^2} \foo + \foo` $x^2 + x^2$ `\futurelet\q\mathrm x\q` xx

`\def\a{1}\xdef\b{\a 2}\b` 12 `\TextOrMath{in text}{in math}` inmath

`\gdef\foo#1{#1^2} \foo{y} + \foo{y}`

$y^2 + y^2$

`\def\f#1{[#1]}\def\g{abc}\expandafter\f\g`

$[a]bc$

`\def\a{1}\edef\b{\a 2}\def\a{9}\b`

12

`\def\a{1}\edef\b{\noexpand\a 2}\def\a{9}\b`

92

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\def\foo{\@ifstar{S}{N}}\foo*\;\foo
\def\foo{\@ifnextchar[{Y}{N}}\foo[x]\;\foo
\@firstoftwo{a}{b}\;\@secondoftwo{a}{b}
```

Operators

Big Operators

`\sum`	$\sum$	`\prod`	$\prod$	`\bigotimes`	$\bigotimes$
`\bigvee`	$\bigvee$	`\int`	$\int$	`\coprod`	$\coprod$
`\bigoplus`	$\bigoplus$	`\bigwedge`	$\bigwedge$	`\iint`	$\iint$
`\intop`	$\int$	`\bigodot`	$\bigodot$	`\bigcap`	$\bigcap$
`\iiint`	$\iiint$	`\smallint`	$\int$	`\biguplus`	$\biguplus$
`\bigcup`	$\bigcup$	`\oint`	$\oint$	`\bigsqcup`	$\bigsqcup$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\oiint` `\oiiint`

Binary Operators

`+`	$+$	`\cdot`	$\cdot$	`\gtrdot`	$\gtrdot$
`x \pmod a`	$x \pmod a$	`-`	$-$	`\cdotp`	$\cdotp$
`\intercal`	$\intercal$	`x \pod a`	$x \pod a$	`/`	$/$
`\centerdot`	$\centerdot$	`\land`	$\land$	`\rhd`	$\rhd$
`*`	$*$	`\circ`	$\circ$	`\leftthreetimes`	$\leftthreetimes$
`\rightthreetimes`	$\rightthreetimes$	`\amalg`	$\amalg$	`\circledast`	$\circledast$
`\ldotp`	$\ldotp$	`\rtimes`	$\rtimes$	`\And`	$\And$
`\circledcirc`	$\circledcirc$	`\lor`	$\lor$	`\setminus`	$\setminus$
`\ast`	$\ast$	`\circleddash`	$\circleddash$	`\lessdot`	$\lessdot$
`\smallsetminus`	$\smallsetminus$	`\barwedge`	$\barwedge$	`\Cup`	$\Cup$
`\lhd`	$\lhd$	`\sqcap`	$\sqcap$	`\bigcirc`	$\bigcirc$
`\cup`	$\cup$	`\ltimes`	$\ltimes$	`\sqcup`	$\sqcup$
`\bmod`	$\bmod$	`\curlyvee`	$\curlyvee$	`x \mod a`	$x \mod a$
`\times`	$\times$	`\boxdot`	$\boxdot$	`\curlywedge`	$\curlywedge$
`\mp`	$\mp$	`\unlhd`	$\unlhd$	`\boxminus`	$\boxminus$
`\div`	$\div$	`\odot`	$\odot$	`\unrhd`	$\unrhd$
`\boxplus`	$\boxplus$	`\divideontimes`	$\divideontimes$	`\ominus`	$\ominus$
`\uplus`	$\uplus$	`\boxtimes`	$\boxtimes$	`\dotplus`	$\dotplus$
`\oplus`	$\oplus$	`\vee`	$\vee$	`\bullet`	$\bullet$
`\doublebarwedge`	$\doublebarwedge$	`\otimes`	$\otimes$	`\veebar`	$\veebar$
`\Cap`	$\Cap$	`\doublecap`	$\Cap$	`\oslash`	$\oslash$
`\wedge`	$\wedge$	`\cap`	$\cap$	`\doublecup`	$\Cup$
`\pm`	$\pm$	`\wr`	$\wr$		

Fractions and Binomials

`\frac{a}{b}`	$\frac{a}{b}$	`\tfrac{a}{b}`	$\tfrac{a}{b}$	`{a \over b}`	$\{a \over b\}$
`\dfrac{a}{b}`	$\dfrac{a}{b}$	`a/b`	a/b	`\binom{n}{k}`	$\binom{n}{k}$
`\dbinom{n}{k}`	$\dbinom{n}{k}$	`{n\brace k}`	$\{n \brace k\}$	`{n \choose k}`	$\{n \choose k\}$
`\tbinom{n}{k}`	$\tbinom{n}{k}$	`{n\brack k}`	$\left[n \brack k\right]$		

`\genfrac ( ] {2pt}{1}a{a+1}` $\left(\genfrac{}{}{2pt}{1}{a}{a+1}\right]$ `\cfrac{a}{1 + \cfrac{1}{b}}` $\cfrac{a}{1+\cfrac{1}{b}}$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

Operators / Fractions and Binomials (continued)

```
{a \above{2pt} b+1}
```

Math Operators

`\arcsin`	$\arcsin$	`\deg`	$\deg$	`\sec`	$\sec$
`\arccos`	$\arccos$	`\cosh`	$\cosh$	`\dim`	$\dim$
`\sin`	$\sin$	`\arctan`	$\arctan$	`\cot`	$\cot$
`\exp`	$\exp$	`\sinh`	$\sinh$	`\hom`	$\hom$
`\coth`	$\coth$	`\ker`	$\ker$	`\tan`	$\tan$
`\arg`	$\arg$	`\csc`	$\csc$	`\lg`	$\lg$
`\tanh`	$\tanh$	`\ln`	$\ln$	`\cos`	$\cos$
`\log`	$\log$	`\operatorname{f}`	f	`\injlim`	$\injlim$
`\min`	$\min$	`\varinjlim`	$\varinjlim$	`\lim`	$\lim$
`\varliminf`	$\varliminf$	`\det`	$\det$	`\liminf`	$\liminf$
`\Pr`	$\Pr$	`\varlimsup`	$\varlimsup$	`\gcd`	$\gcd$
`\limsup`	$\limsup$	`\projlim`	$\projlim$	`\varprojlim`	$\varprojlim$
`\inf`	$\inf$	`\max`	$\max$	`\sup`	$\sup$

`\operatorname*{f}` $\operatorname*{f}$

`\operatornamewithlimits{f}` $\operatornamewithlimits{f}$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\cosec` `\arctg` `\cotg`
`\sh` `\arcctg` `\ch`
`\ctg` `\tg` `\cth`
`\th` `\argmax` `\argmin`
`\plim`

Square roots

`\sqrt{x}` $\sqrt{x}$ `\sqrt[3]{x}` $\sqrt[3]{x}$

Relations

`\stackrel{!}{=}`	$\stackrel{!}{=}$	`=`	$=$	`\doteqdot`	$\doteqdot$
`\lessapprox`	$\lessapprox$	`\smile`	$\smile$	`<`	$<$
`\eqcirc`	$\eqcirc$	`\lesseqgtr`	$\lesseqgtr$	`\sqsubset`	$\sqsubset$
`>`	$>$	`\eqcolon`	$\eqcolon$	`\lesseqqgtr`	$\lesseqqgtr$
`\sqsubseteq`	$\sqsubseteq$	`:`	$:$	`\Eqcolon`	$\Eqcolon$
`\lessgtr`	$\lessgtr$	`\sqsupset`	$\sqsupset$	`\approx`	$\approx$
`\eqqcolon`	$\eqqcolon$	`\lesssim`	$\lesssim$	`\sqsupseteq`	$\sqsupseteq$
`\approxcolon`	$\approxcolon$	`\Eqqcolon`	$\Eqqcolon$	`\ll`	$\ll$
`\Subset`	$\Subset$	`\eqsim`	$\eqsim$	`\lll`	$\lll$
`\subset`	$\subset$	`\approxeq`	$\approxeq$	`\eqslantgtr`	$\eqslantgtr$
`\llless`	$\llless$	`\subseteq`	$\subseteq$	`\asymp`	$\asymp$
`\eqslantless`	$\eqslantless$	`\subseteqq`	$\subseteqq$	`\backepsilon`	$\backepsilon$
`\equiv`	$\equiv$	`\mid`	$\mid$	`\succ`	$\succ$
`\backsim`	$\backsim$	`\fallingdotseq`	$\fallingdotseq$	`\models`	$\models$
`\succapprox`	$\succapprox$	`\backsimeq`	$\backsimeq$	`\frown`	$\frown$
`\multimap`	$\multimap$	`\succcurlyeq`	$\succcurlyeq$	`\between`	$\between$
`\ge`	$\ge$	`\succeq`	$\succeq$	`\bowtie`	$\bowtie$
`\geq`	$\geq$	`\owns`	$\owns$	`\succsim`	$\succsim$

Relations (continued)

\bumpeq	≏	\geqq	≧	\parallel	∥
\Supset	⋑	\Bumpeq	≎	\geqslant	⩾
\perp	⊥	\supset	⊃	\circeq	≗
\gg	≫	\pitchfork	⋔	\supseteq	⊇
\colonapprox	:≈	\ggg	⋙	\prec	≺
\supseteqq	⫆	\Colonapprox	::≈	\gggtr	⋙
\precapprox	⪷	\thickapprox	≈	\coloneq	≔
\preccurlyeq	≼	\thicksim	∼	\Coloneq	⩴
\gtrapprox	⪆	\preceq	⪯	\trianglelefteq	⊴
\coloneqq	≔	\gtreqless	⋚	\precsim	≾
\triangleq	≜	\Coloneqq	⩴	\gtreqqless	⪌
\propto	∝	\trianglerighteq	⊵	\colonsim	:∼
\gtrless	≷	\risingdotseq	≓	\varpropto	∝
\Colonsim	::∼	\gtrsim	≳	\shortmid	∣
\vartriangle	△	\cong	≅	\shortparallel	∥
\vartriangleleft	◁	\curlyeqprec	⋞	\in	∈
\sim	∼	\vartriangleright	▷	\curlyeqsucc	⋟
\Join	⋈	\simcolon	∼:	\vcentcolon	:
\dashv	⊣	\le	≤	\vdash	⊢
\dblcolon	::	\leq	≤	\simeq	≃
\vDash	⊨	\doteq	≐	\leqq	≦
\smallfrown	⌢	\Vdash	⊩	\Doteq	≑
\leqslant	⩽	\smallsmile	⌣	\Vvdash	⊪

Drawn by the package; LaTeX has no equivalent, so no output is shown:

\approxcoloncolon \lt \origof
\gt \imageof \simcoloncolon

Negated Relations

\not =	≠	\gnapprox	⪊	\ngeqslant	⩾̸
\nsubseteq	⊈	\precneqq	⪵	\gneq	⪈
\ngtr	≯	\nsubseteqq	⫅̸	\precnsim	⋨
\gneqq	≩	\nleq	≰	\nsucc	⊁
\subsetneq	⊊	\gnsim	⋧	\nleqq	≦̸
\nsucceq	⋡	\subsetneqq	⫋	\gvertneqq	≩
\nleqslant	⩽̸	\nsupseteq	⊉	\succnapprox	⪺
\lnapprox	⪉	\nless	≮	\nsupseteqq	⫆̸
\succneqq	⪶	\lneq	⪇	\nmid	∤
\ntriangleleft	⋪	\succnsim	⋩	\lneqq	≨
\notin	∉	\ntrianglelefteq	⋬	\supsetneq	⊋
\lnsim	⋦	\ntriangleright	⋫	\supsetneqq	⫌
\lvertneqq	≨	\nparallel	∦	\ntrianglerighteq	⋭
\varsubsetneq	⊊	\ncong	≇	\nprec	⊀
\nvdash	⊬	\varsubsetneqq	⫋	\ne	≠
\npreceq	⋠	\nvDash	⊭	\varsupsetneq	⊋
\neq	≠	\nshortmid	∤	\nVDash	⊯

Relations / Negated Relations (continued)

\varsupsetneqq	$\varsupsetneqq$	\ngeq	$\ngeq$	\nshortparallel	$\nshortparallel$
\nVdash	$\nVdash$	\ngeqq	$\ngeqq$	\nsim	$\nsim$
\precnapprox	$\precnapprox$				

Drawn by the package; LaTeX has no equivalent, so no output is shown:

\notni

Arrows

\circlearrowleft	$\circlearrowleft$	\leftharpoonup	$\leftharpoonup$	\circlearrowright	$\circlearrowright$
\leftleftarrows	$\leftleftarrows$	\curvearrowleft	$\curvearrowleft$	\leftrightarrow	$\leftrightarrow$
\restriction	$\restriction$	\curvearrowright	$\curvearrowright$	\Leftrightarrow	$\Leftrightarrow$
\rightarrow	$\rightarrow$	\leftrightarrows	$\leftrightarrows$	\Rightarrow	$\Rightarrow$
\rightarrowtail	$\rightarrowtail$	\rightharpoondown	$\rightharpoondown$	\dashleftarrow	$\dashleftarrow$
\Lleftarrow	$\Lleftarrow$	\rightharpoonup	$\rightharpoonup$	\dashrightarrow	$\dashrightarrow$
\longleftarrow	$\longleftarrow$	\rightleftarrows	$\rightleftarrows$	\downarrow	$\downarrow$
\Longleftarrow	$\Longleftarrow$	\Downarrow	$\Downarrow$	\rightrightarrows	$\rightrightarrows$
\downdownarrows	$\downdownarrows$	\rightsquigarrow	$\rightsquigarrow$	\downharpoonleft	$\downharpoonleft$
\longmapsto	$\longmapsto$	\Rrightarrow	$\Rrightarrow$	\downharpoonright	$\downharpoonright$
\longrightarrow	$\longrightarrow$	\Rsh	$\Rsh$	\gets	$\gets$
\Longrightarrow	$\Longrightarrow$	\searrow	$\searrow$	\looparrowleft	$\looparrowleft$
\swarrow	$\swarrow$	\looparrowright	$\looparrowright$	\to	$\to$
\twoheadleftarrow	$\twoheadleftarrow$	\hookleftarrow	$\hookleftarrow$	\hookrightarrow	$\hookrightarrow$
\iff	$\iff$	\Lsh	$\Lsh$	\impliedby	$\impliedby$
\mapsto	$\mapsto$	\implies	$\implies$	\nearrow	$\nearrow$
\uparrow	$\uparrow$	\nleftarrow	$\nleftarrow$	\Uparrow	$\Uparrow$
\nLeftarrow	$\nLeftarrow$	\updownarrow	$\updownarrow$	\nleftrightarrow	$\nleftrightarrow$
\Updownarrow	$\Updownarrow$	\leadsto	$\leadsto$	\nLeftrightarrow	$\nLeftrightarrow$
\upharpoonleft	$\upharpoonleft$	\leftarrow	$\leftarrow$	\nrightarrow	$\nrightarrow$
\upharpoonright	$\upharpoonright$	\Leftarrow	$\Leftarrow$	\nRightarrow	$\nRightarrow$
\upuparrows	$\upuparrows$	\leftarrowtail	$\leftarrowtail$	\nwarrow	$\nwarrow$
\mapsfrom	$\mapsfrom$	\leftharpoondown	$\leftharpoondown$	\xleftarrow{abc}	$\xleftarrow{abc}$
\xLeftarrow{abc}	$\xLeftarrow{abc}$	\xRightarrow{abc}	$\xRightarrow{abc}$	\xmapsto{abc}	$\xmapsto{abc}$

\leftrightharpoons	$\leftrightharpoons$	\leftrightsquigarrow	$\leftrightsquigarrow$
\rightleftharpoons	$\rightleftharpoons$	\longleftrightarrow	$\longleftrightarrow$
\Longleftrightarrow	$\Longleftrightarrow$	\twoheadrightarrow	$\twoheadrightarrow$
\xrightarrow[under]{over}	$\xrightarrow[under]{over}$	\xleftrightarrow{abc}	$\xleftrightarrow{abc}$
\xLeftrightarrow{abc}	$\xLeftrightarrow{abc}$	\xhookleftarrow{abc}	$\xhookleftarrow{abc}$
\xhookrightarrow{abc}	$\xhookrightarrow{abc}$	\xleftharpoonup{abc}	$\xleftharpoonup{abc}$
\xrightharpoonup{abc}	$\xrightharpoonup{abc}$	\xleftharpoondown{abc}	$\xleftharpoondown{abc}$
\xrightharpoondown{abc}	$\xrightharpoondown{abc}$	\xleftrightharpoons{abc}	$\xleftrightharpoons{abc}$
\xrightleftharpoons{abc}	$\xrightleftharpoons{abc}$		

Drawn by the package; LaTeX has no equivalent, so no output is shown:

\rArr	\rarr	\Darr
\dArr	\darr	\Harr
\hArr	\harr	\Lrarr
\lrArr	\lrarr	\Uarr
\uArr	\uarr	\Larr
\lArr	\larr	\Rarr

Relations / Arrows (continued)

```
\xlongequal{abc}          \xtofrom{abc}    \xtwoheadrightarrow{abc}
\xtwoheadleftarrow{abc}
```

Special Notation

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\bra{\phi}   \ket{\psi}   \braket{\phi\VERT\psi}
\Bra{\phi}   \Ket{\psi}
```

Style, Color, Size, and Font

`\color{blue} F=ma`	$F = ma$	`\mathrm{Ab0}`	Ab0	`\mathbf{Ab0}`	**Ab0**
`\mathsf{Ab0}`	Ab0	`\mathnormal{Ab0}`	*Ab0*	`\textbf{Ab0}`	**Ab0**
`\textsf{Ab0}`	Ab0	`\textrm{Ab0}`	Ab0	`\bf Ab0`	**Ab0**
`\sf Ab0`	Ab0	`\rm Ab0`	Ab0	`\bold{Ab0}`	**Ab0**
`\textnormal{Ab0}`	Ab0	`\boldsymbol{Ab0}`	**Ab0**	`\Bbb{AB}`	$\mathbb{AB}$
`\text{Ab0}`	Ab0	`\bm{Ab0}`	**Ab0**	`\mathbb{AB}`	$\mathbb{AB}$
`\textup{Ab0}`	Ab0	`\textmd{Ab0}`	Ab0	`\frak{Ab0}`	$\mathfrak{Ab0}$
`\mathit{Ab0}`	*Ab0*	`\mathtt{Ab0}`	Ab0	`\mathfrak{Ab0}`	$\mathfrak{Ab0}$
`\textit{Ab0}`	*Ab0*	`\texttt{Ab0}`	Ab0	`\mathcal{AB0}`	$\mathcal{AB0}$
`\it Ab0`	*Ab0*	`\tt Ab0`	Ab0	`\cal AB0`	$\mathcal{AB0}$
`\emph{Ab0}`	*Ab0*	`\mathscr{AB}`	$\mathscr{AB}$	`\scriptstyle x`	x
`\lim\limits_x`	$\lim_x$	`\lim\nolimits_x`	$\lim_x$	`\verb!x^2!`	x^2

`\textcolor{blue}{F=ma}` $F = ma$ `\textsf{\textbf{H}}` **H**

`\mathsf{\mathbf{H}}` **H** `\displaystyle\sum_{i=1}^n` $\displaystyle\sum_{i=1}^{n}$

`\textstyle\sum_{i=1}^n` $\textstyle\sum_{i=1}^{n}$ `\scriptscriptstyle x` x

`\mathchoice{D}{T}{S}{SS}` T

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\textcolor{#228B22}{F=ma}   \colorbox{aqua}{$F=ma$}   \mathsfit{Ab0}
\pmb{\mu}                   \Huge AB                  \normalsize AB
\huge AB                    \small AB                 \LARGE AB
\footnotesize AB            \Large AB                 \scriptsize AB
\large AB                   \tiny AB                  \sixptsize sixptsize
\fcolorbox{red}{aqua}{$F=ma$}
```

Symbols and Punctuation

`\dots`	…	`\%`	%	`\cdots`	⋯
`\#`	#	`\ddots`	⋱	`\&`	&
`\ldots`	…	`\nabla`	∇	`\_`	_
`\vdots`	⋮	`\infty`	∞	`\dotsb`	⋯
`\text{--}`	–	`\dotsc`	…	`\checkmark`	✓
`\dotsi`	⋯	`\dag`	†	`\text{---}`	—
`\dotsm`	⋯	`\dagger`	†	`\dotso`	…
`\ddag`	‡	`\mathellipsis`	…	`\ddagger`	‡
`` ` ``	`	`\Box`	□	`\lq`	‘
`\square`	□	`\angle`	∠	`\blacksquare`	■
`\measuredangle`	∡	`\rq`	’	`\triangle`	△
`\sphericalangle`	∢	`\triangledown`	▽	`\top`	⊤
`"`	"	`\triangleleft`	◁	`\bot`	⊥

Symbols and Punctuation (continued)

\triangleright	▷	\$	$	\colon	:
\bigtriangledown	▽	\backprime	‵	\bigtriangleup	△
\pounds	£	\prime	′	\blacktriangle	▲
\mathsterling	£	\text{\textless}	<	\yen	¥
\text{\textbar}	|	\surd	√	\diamond	⋄
\Diamond	◇	\lozenge	◊	\mho	℧
\blacklozenge	⧫	\diagdown	╲	\text{\P}	¶
\star	★	\diagup	╱	\text{\S}	§
\bigstar	★	\flat	♭	\clubsuit	♣
\natural	♮	\copyright	©	\sharp	♯
\circledR	®	\diamondsuit	♢	\heartsuit	♡
\circledS	Ⓢ	\spadesuit	♠	\maltese	✠
\minuso	⦵				

\text{\textunderscore}	_	\text{\textendash}	–
\text{\textemdash}	—	\text{\textdagger}	†
\text{\textasciitilde}	~	\text{\textasciicircum}	^
\text{\textellipsis}	…	\text{\textdaggerdbl}	‡
\text{\textquoteleft}	‘	\text{\textquoteright}	’
\text{\textquotedblleft}	“	\text{\textquotedblright}	”
\text{\textdollar}	$	\blacktriangledown	▼
\text{\textsterling}	£	\text{\textgreater}	>
\blacktriangleleft	◀	\blacktriangleright	▶
\text{\textbardbl}	‖	\text{\textbraceleft}	{
\text{\textdegree}	°	\text{\textbraceright}	}
\text{\textbackslash}	\	\text{\textregistered}	®

Drawn by the package; LaTeX has no equivalent, so no output is shown:

\KaTeX	\LaTeX	\TeX
\infin	\sdot	\Dagger
\degree	\text{\sect}	\clubs
\diamonds	\hearts	\spades

Units

\rule{1em}{1em} ■ \rule{10pt}{10pt} ■

Drawn by the package; LaTeX has no equivalent, so no output is shown:

```
\huge\rule{1em}{1em}     \rule{18mu}{18mu}    \huge\rule{18mu}{18mu}
\huge\rule{10pt}{10pt}
\scriptscriptstyle\rule{1em}{1em}
\scriptscriptstyle\rule{18mu}{18mu}
\scriptscriptstyle\rule{10pt}{10pt}
```

Extensions

mhchem

`\ce{H2O}` $\ce{H2O}$ `\ce{1/2 H2O}` $\ce{1/2 H2O}$ `\ce{CuSO4*5H2O}` $\ce{CuSO4*5H2O}$

`\ce{CO2 + C -> 2 CO}` $\ce{CO2 + C -> 2 CO}$ `\ce{^{227}_{90}Th+}` $\ce{^{227}_{90}Th+}$

`\ce{Fe^{3+} + 3e- -> Fe(s)}` $\ce{Fe^{3+} + 3e- -> Fe(s)}$

`\ce{SO4^2- + Ba^2+ -> BaSO4 v}`

$\ce{SO4^2- + Ba^2+ -> BaSO4 v}$

`\ce{Zn^2+ <=>[+ 2OH-][+ 2H+] Zn(OH)2 v}`

$\ce{Zn^2+ <=>[+ 2OH-][+ 2H+] Zn(OH)2 v}$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\pu{1.5 mol/L}` `\pu{123 kJ/mol}` `\pu{1.2e-3 mol L-1}`

Further LaTeX

More commands

`\idotsint` $\idotsint$ `\iiiint` $\iiiint$

`\text{\textsection\textparagraph\textcopyright\texttrademark}`

$\text{\textsection\textparagraph\textcopyright\texttrademark}$

Drawn by the package; LaTeX has no equivalent, so no output is shown:

`\euro`

Numbers and units (a subset of siunitx)

`\num{12345.6789}` $\num{12345.6789}$ `\num{1.5e-4}` $\num{1.5e-4}$ `\num{1.2(3)e3}` $\num{1.2(3)e3}$

`\si{kg.m.s^-2}` $\si{kg.m.s^-2}$ `\ang{1;2;3}` $\ang{1;2;3}$ `\numrange{1}{10}` $\numrange{1}{10}$

`\numlist{1;2;3}` $\numlist{1;2;3}$

`\qty{1.5}{\kilo\gram}` $\qty{1.5}{\kilo\gram}$ `\SIrange{1}{5}{\volt}` $\SIrange{1}{5}{\volt}$

`\SI{9.81}{\meter\per\second\squared}`

$\SI{9.81}{\meter\per\second\squared}$

Equations

`\cancelto{0}{x}` $\cancelto{0}{x}$

`\bigl\lvert x \bigr\rvert` $\bigl\lvert x \bigr\rvert$

`\begin{dcases} x & x>0 \\ -x & \text{otherwise} \end{dcases}`

$$\begin{dcases} x & x>0 \\ -x & \text{otherwise} \end{dcases}$$

`\begin{subarray}{c} i<n \\ j<m \end{subarray}`

$\begin{subarray}{c} i<n \\ j<m \end{subarray}$

`\sqrt[\leftroot{-2}\uproot{2}n]{x}`

$\sqrt[\leftroot{-2}\uproot{2}n]{x}$

Further LaTeX / Equations (continued)

```
a = b \tag{1}
```

$$a = b \tag{1}$$

```
   a = b \tag*{(*)}
```

$$a = b \tag{*}$$

Tables

```
\begin{tabular}{|l|r|} \hline a & 1 \\ \hline b & 22 \\ \hline
\end{tabular}
```

a	1
b	22

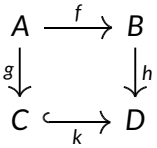
```
\begin{array}{l} \textbf{Bold} \\ \textsc{Small caps} \\
\verb|verbatim| \end{array}
```

Bold
SMALL CAPS
`verbatim`

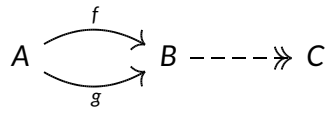
Drawn by the package; LaTeX has no equivalent, so no output is shown:
`\begin{array}{c:c} a & b \\ \hdashline c & d \end{array}`

Commutative diagrams

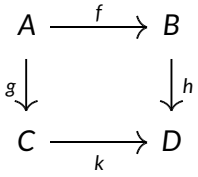
```
\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "g"'] & B \arrow[d,
"h"] \\ C \arrow[r, hook, "k"'] & D \end{tikzcd}
```




```
\begin{tikzcd}[column sep=large] A \arrow[r, "f", bend left]
\arrow[r, "g"', bend right] & B \arrow[r, two heads, dashed] & C
\end{tikzcd}
```

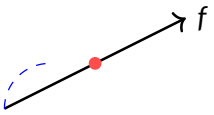



```
\begin{CD} A @>f>> B \\ @VgVV @VVhV \\ C @>>k> D \end{CD}
```

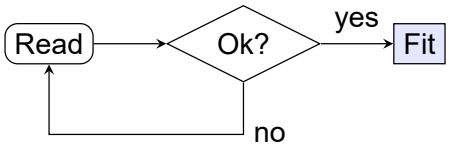


TikZ pictures

```
\begin{tikzpicture}\draw[->, thick] (0,0) -- (2,1) node[right]
{$f$}; \fill[red!70] (1,0.5) circle (2pt); \draw[blue, dashed]
(0,0) arc (180:90:0.5);\end{tikzpicture}
```



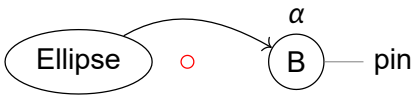

```
\begin{tikzpicture}[node distance=8mm, >=stealth] \node[draw,
rounded corners] (a) {Read}; \node[draw, diamond, aspect=2,
right=of a] (b) {Ok?}; \node[draw, fill=blue!10, right=of b] (c)
{Fit}; \draw[->] (a) -- (b); \draw[->] (b) -- node[above] {yes}
(c); \draw[->] (b) |- ++(0,-1) node[right] {no} -|
(a);\end{tikzpicture}
```



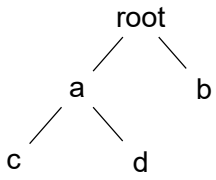

```
\begin{tikzpicture}\foreach \x/\c in
{0/red,1/orange,2/green,3/blue} {\fill[\c] (\x*0.5,0) circle
(0.2);}\end{tikzpicture}
```



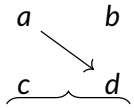
```
\begin{tikzpicture}\node[draw, ellipse] (a) at (0,0) {Ellipse};
\node[draw, circle, label=above:{$\alpha$}, pin=right:pin] (b)
at (2.4,0) {B}; \draw[->] (a) to[bend left] (b); \draw[red]
($(a)!0.5!(b)$) circle (2pt);\end{tikzpicture}
```




```
\begin{tikzpicture}[level distance=8mm, sibling distance=14mm]
\node {root} child {node {a} child {node {c}} child {node {d}}}
child {node {b}};\end{tikzpicture}
```




```
\begin{tikzpicture}\matrix (m) [matrix of nodes, row sep=3mm,
column sep=6mm] {$a$ & $b$ \\ $c$ & $d$ \\}; \draw[->] (m-1-1)
-- (m-2-2); \draw[decorate, decoration={brace, amplitude=4pt}]
(m-2-1.south west) -- (m-2-2.south east);\end{tikzpicture}
```



Boxes and columns

```
\begin{tcolorbox}[title=A box] Text in a box, which is as wide
as the line. \end{tcolorbox}
```

A box

Text in a box, which is as wide as the line.

```
   \begin{tcolorbox}[colback=yellow!15, colframe=blue!50!black,
colbacktitle=blue!15, coltitle=black, title=Colours] A title
band, a frame, and a body in colours of its own. \end{tcolorbox}
```

Colours

A title band, a frame, and a body in colours of its own.

```
   \begin{multicols}{2}[A header over both columns] One paragraph
of some words that run on to a second line.\par Another, in the
first column too.\par A third, which is balanced into the
second.\par A fourth. \end{multicols}
```

A header over both columns

One paragraph of some words that run on to a second line.
Another, in the first column too.

A third, which is balanced into the second.
A fourth.